

pharmaniaga[®]

Potassium Chloride

10% w/v Injection



COMPOSITION

Each ampoule contains Potassium Chloride 100mg/mL.

DESCRIPTION

A colourless or pale yellow solution. Single dose container

PHARMACODYNAMICS

Potassium is the predominant cation (150 mEq/L) within cells. Intracellular sodium content is relatively low. In extracellular fluid, sodium predominates and the potassium content is low (4 to 5 mEq/L).

A membrane-bound enzyme, sodium-potassium activated adenosine triphosphatase ($\text{Na}^+ + \text{K}^+ + \text{ATPase}$), actively transports or pumps sodium out of and potassium into cells in order to maintain these concentration gradients. The gradients are necessary for the conduction of nerve impulses in such specialised tissues as the heart, brain and skeletal muscle and to maintain normal renal function and acid-base balance. High intracellular concentrations are necessary for numerous cellular metabolic processes.

PHARMACOKINETICS

Potassium is mainly excreted in the urine (90%). Only a small amount is excreted in faeces (10%).

INDICATION

Hypokalemia (Treatment) -Potassium Chloride injection is indicated in patients with hypokalaemia with or without metabolic alkalosis, in digitalis intoxication, and in patients with hypokalaemia familial periodic paralysis.

RECOMMENDED DOSAGE

Note: 1 gm of potassium chloride provides 13.41 mEq of potassium.

**FOR SLOW IV INFUSION
AFTER DILUTION.
NOT FOR IV BOLUS**

Fluid balance, serum electrolytes and acid-base balance may need to be monitored before and during administration, with particular attention to serum sodium in patients with increased non-osmotic vasopressin release (syndrome of inappropriate antidiuretic hormone secretion, SIADH) and in patients co-medicated with vasopressin agonist drugs, due to the risk of hospital acquired hyponatraemia (see sections Warning and Precautions, Interaction with Other Medicaments and Side Effects)

Monitoring of serum sodium is particularly important for hypotonic fluids.

The infusion rate and volume depend on the age, weight, clinical condition (e.g. burns, surgery, head injury, infections), and concomitant therapy should be determined by the consulting physician experienced in paediatric intravenous fluid therapy (see sections Warning and Precautions and Side Effects).

Adult Dose:

The dose rate of infusion must be determined by the individual requirements. The response of the patients as determined by serum potassium concentration and the electrocardiogram following the initial 40 to 60 mEq infused should indicate the subsequent infusion rate required.

Paediatric Dose:

Intravenous infusion, up to the equivalent of 3 mEq of potassium per kg body weight per day.

ROUTE OF ADMINISTRATION

Parenteral

By slow intravenous infusion as dilute solution.

CONTRAINDICATIONS

Administration of potassium chloride injections is contraindicated when signs of hyperkalaemia are present. They should be administered with careful

patient monitoring when the following medical problems such as acidosis, cardiac disease, chronic renal insufficiency, Addison's disease or hypoadrenalism exist. Severe or prolonged diarrhoea leading to dehydration, acute dehydration, trauma-related medical problems eg. severe burns, extensive infection, crush injuries, massive haemolysis or major surgery in the preceding 24 hours.

WARNINGS AND PRECAUTIONS

During intravenous potassium chloride administration to avoid hyperkalaemia, the infusion rate must not be rapid; a rate of 10 mEq of potassium per hour is usually considered to be safe as long as urine output is adequate. As a general rule, the rate should never exceed 1 mEq per minute for adults, or 0.02 mEq per kg body weight per minute for children.

High volume infusion must be used under specific monitoring in patients with cardiac or pulmonary failure, and in patients with non-osmotic vasopressin release (including SIADH), due to the risk of hospital-acquired hyponatraemia (see below).

Hyponatraemia

Patients with non-osmotic vasopressin release (e.g. in acute illness, pain, post-operative stress, infections, burns, and CNS diseases), patients with heart-, liver- and kidney diseases and patients exposed to vasopressin agonists (see section Interaction with Other Medicaments) are at particular risk of acute hyponatraemia upon infusion of hypotonic fluids.

Acute hyponatraemia can lead to acute hyponatraemic encephalopathy (cerebral oedema) characterized by headache, nausea, seizures, lethargy and vomiting. Patients with cerebral oedema are at particular risk of severe, irreversible and life-threatening brain injury.

Children, women in the fertile age and patients with reduced cerebral compliance (e.g. meningitis, intracranial bleeding, cerebral contusion and brain oedema) are at particular risk of the severe and life-threatening brain swelling caused by acute hyponatraemia.

INTERACTION WITH OTHER MEDICAMENTS

Combination containing any of the following medications, depending on the amount present, may also interact with this medication. Adrenocorticoids, mineralocorticoid, corticotropin (ACTH), anti inflammatory analgesics, nonsteroidal (NSAIDs),

antimuscarinics or other medications with antimuscarinic activity, potassium-sparing diuretics (amiloride, spironolactone, or triamterene), potassium containing medications, captopril, enalapril, lisinopril, digitalis glycosides in the presence of heart block, heparin, laxatives, quinidine, vitamin B12.

The below listed drugs increase the vasopressin effect, leading to reduced renal electrolyte free water excretion and may increase the risk of hospital acquired hyponatraemia following inappropriately balanced treatment with i.v. fluids (see sections Recommended Dosage, Warning and Precautions and Side Effects).

- Drugs stimulating vasopressin release include:
Chlorpropamide, clofibrate, carbamazepine, vincristine, selective serotonin reuptake inhibitors, 3,4-methylenedioxy-N- methamphetamine, ifosfamide, antipsychotics, narcotics.
- Drugs potentiating vasopressin action include:
Chlorpropamide, NSAIDs, cyclophosphamide
- Vasopressin analogues include:
Desmopressin, oxytocin, vasopressin, terlipressin

Other medicinal products increasing the risk of hyponatraemia also include diuretics in general and antiepileptics such as oxcarbazepine.

Incompatibilities

Potassium chloride concentrates for injection have been reported to be physically incompatible with IV solutions containing various drugs. The compatibility of the large volume IV fluid intended for dilution should be checked before use.

STATEMENT ON USAGE DURING PREGNANCY AND LACTATION

Pharmaniaga Potassium Chloride 10% W/V Injection should be administered with special caution for pregnant women during labour particularly as to serum-sodium if administered in combination with oxytocin (see section Warning and Precautions, Interaction with Other Medicaments and Side Effects). Pharmaniaga Potassium chloride 10% W/V Injection may be used during pregnancy and lactation under the supervision of the prescribing physician if considered essential by the physician.

SIDE EFFECTS

Excessive doses of potassium may lead to the development of hyperkalaemia, especially in patients with renal impairment. Symptoms may include paraesthesia of the extremities, muscle weakness, paralysis, cardiac arrhythmias, heart block, cardiac arrest and confusion. Cardiac toxicity is of particular concern after intravenous dosage. Pain or phlebitis may occur when given intravenously via peripheral veins particularly at higher concentrations.

- Hospital acquired hyponatraemia*
- Acute hyponatraemic encephalopathy*

*Hospital acquired hyponatraemia may cause irreversible brain injury and death, due to development of acute hyponatraemic encephalopathy, frequency unknown (see sections Recommended Dosage, Warning and Precautions and Interaction with Other Medicaments).

SYMPTOMS AND TREATMENT OF OVERDOSAGE

Symptoms

Toxic doses commonly cause hyperkalaemia with paraesthesia of the extremities, listlessness, mental confusion, weakness, paralysis, hypotension, cardiac arrhythmias, heart block, and cardiac arrest.

Treatment

Treatment of hyperkalaemia includes discontinuing food and medication which contain potassium and any potassium sparing diuretics, intravenous administration of insulin in 10% to 25% dextrose injection (using 10 units of insulin per 20 grams of dextrose) at a rate of 300 to 500 mL of solution per hour; correction of any existing acidosis with intravenous sodium bicarbonate; and utilisation of exchange resin, haemodialysis or peritoneal dialysis. Administer a calcium salt (calcium gluconate 0.5 to 1.0 g over 2 minutes) to antagonise the cardiotoxic effects in the patients whose ECG show absent P waves or broad QRS complex and who are not receiving digitalis glycosides (may be repeated after 2 minute intervals).

STORAGE CONDITION

Store below 30°C.

Caution: Do not use if solution contains growth, particles, turbidity, or if there is any change of appearance. The compatibility of large volume IV fluid intended for dilution should be checked before use.

SHELF LIFE

3 years from date of manufacture.

DOSAGE FORMS AND PACKAGING

AVAILABLE:

10 x 10mL ampoule (clear).

PRODUCT REGISTRATION

HOLDER/MANUFACTURER:

PHARMANIAGA LIFESCIENCE SDN. BHD.
(198201002939)

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2002041-V

Revision Date: 24-May-2023